

This assignment will not be graded and is only for practice.

Level 1

1) Consider a function $f : \mathbb{R}^3 \rightarrow \mathbb{R}$.

1 point

Choose the correct statement with respect to ∇f .

- ☐ ∇f has co-domain $\mathbb{R}^2$
- ☐ ∇f has co-domain $\mathbb{R}$
- ☐ ∇f has co-domain $\mathbb{R}^3$
- ☐ ∇f has domain $\mathbb{R}^3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

∇f has co-domain $\mathbb{R}^3$

∇f has domain $\mathbb{R}^3$

2) Let $f : \mathbb{R}^3 \rightarrow \mathbb{R}$ and suppose its gradient ∇f is continuous. Which of the following statements is true for the directional derivative of f in the direction of $a \in \mathbb{R}^3$? Consider dot product to be the inner product $\langle, \rangle$. **1 point**

- ☐ $D_a f = \frac{1}{\|a\|} \langle a, \nabla f \rangle$
- ☐ $D_a f = \langle a, \nabla f \rangle$
- ☐ $D_a f = \frac{1}{2\|a\|} \langle a, \nabla f \rangle$
- ☐ $D_a f = \frac{1}{\|a\|^2} \langle a, \nabla f \rangle$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$D_a f = \frac{1}{\|a\|} \langle a, \nabla f \rangle$

3) Let $f(x, y) = e^x \sin y$. If ∇f at the point $(\ln(2), \pi/4)$ is (k, k) where $k \in \mathbb{R}$, then find the value of k^2 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Level 2

4) Let $p = -\frac{k}{(\|r\|)^3}r$, where $r = (x, y, z) \in \mathbb{R}^3 \setminus \{(0, 0, 0)\}$, $k \in \mathbb{R}$.

1 point

Amongst the options below, choose those for which the function f satisfies $\nabla f = p$ for some $k \in \mathbb{R}$.

Consider dot product as the inner product for finding the norm.

☐ $f(x, y, z) = \frac{1}{\|r\|}$

☐ $f(x, y, z) = \frac{1}{2\|r\|}$

☐ $f(x, y, z) = \frac{1}{\|r\|+2}$

☐ $f(x, y, z) = \|r\|$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$f(x, y, z) = \frac{1}{\|r\|}$

$f(x, y, z) = \frac{1}{2\|r\|}$

5) Let $f(x, y, z) = 4(x^2 + y^2) - z^2$.

1 point

Choose the correct option(s).

☐ ∇f at $(1, 0, 2)$ is $(8, 0, -4)$

☐ ∇f is continuous everywhere in $\mathbb{R}^3$

☐ ∇f is not continuous everywhere in $\mathbb{R}^3$

☐ ∇f at $(1, 0, 2)$ is $(-8, 0, -4)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

∇f at $(1, 0, 2)$ is $(8, 0, -4)$

∇f is continuous everywhere in $\mathbb{R}^3$

6) Consider a function defined as

1 point

$$f(x, y) = \begin{cases} \frac{y^3}{x^2+y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Choose the correct option(s).

☐ $\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2+y^2)^2}$, for $(x, y) \neq (0, 0)$

☐ $\frac{\partial f}{\partial x} = \frac{2xy^3}{(x^2+y^2)^2}$, for $(x, y) \neq (0, 0)$

☐ $\frac{\partial f}{\partial y} = \frac{3x^2y^2+y^4}{(x^2+y^2)^2}$ for $(x, y) \neq (0, 0)$

☐ $\frac{\partial f}{\partial y} = \frac{3x^2y^2-y^4}{(x^2+y^2)^2}$ for $(x, y) \neq (0, 0)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2+y^2)^2}, \text{ for } (x, y) \neq (0, 0)$$

$$\frac{\partial f}{\partial y} = \frac{3x^2y^2+y^4}{(x^2+y^2)^2} \text{ for } (x, y) \neq (0, 0)$$

7) Consider the function definition given in Q6 above. Choose the correct option(s).

1 point

☐ $\frac{\partial f}{\partial x}$ is continuous at $(0, 0)$

☐ $\frac{\partial f}{\partial x}$ is not continuous at $(0, 0)$

☐ $\frac{\partial f}{\partial y}$ is not continuous at $(0, 0)$

☐ ∇f is discontinuous at the origin $(0, 0)$

☐ Directional derivative of f in the direction of a vector $a \in \mathbb{R}^2$ at $(0, 0)$ can be obtained using the gradient of f

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{\partial f}{\partial x}$ is not continuous at $(0, 0)$

$\frac{\partial f}{\partial y}$ is not continuous at $(0, 0)$

∇f is discontinuous at the origin $(0, 0)$

8) Consider a function defined as

1 point

$$f(x, y) = \begin{cases} (x^2 + y^2) \sin\left(\frac{1}{\sqrt{x^2 + y^2}}\right) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Choose the correct option(s).

☐ $\frac{\partial f}{\partial y}(0, 0) = 0$

☐ $\frac{\partial f}{\partial x}(0, 0) = 0$

☐ $\frac{\partial f}{\partial x} = 0$

☐ $\frac{\partial f}{\partial y} = 0$

☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{\partial f}{\partial y}(0, 0) = 0$

$\frac{\partial f}{\partial x}(0, 0) = 0$

9) Consider the function definition given in Q8 above. Choose the correct option(s) **1 point**
with respect to the partial derivative of f . Note $\text{sign}(x) = \pm 1$ depending on the sign of x .

☐ $\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \text{sign}(x) \cos\left(\frac{1}{|x|}\right)$ for $x \neq 0$

☐ $\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \text{sign}(x) \cos\left(\frac{1}{|x|}\right)$

- ☐ $\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \cos\left(\frac{1}{|x|}\right)$ for $x = 0$
- ☐ $\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \text{sign}(y) \cos\left(\frac{1}{|y|}\right)$ for $y = 0$
- ☐ $\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \text{sign}(y) \cos\left(\frac{1}{|y|}\right)$
- ☐ $\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \cos\left(\frac{1}{|y|}\right)$ for $y = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \text{sign}(x) \cos\left(\frac{1}{|x|}\right) \text{ for } x = 0$$

$$\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \text{sign}(y) \cos\left(\frac{1}{|y|}\right) \text{ for } y = 0$$

10) Consider the function definition given in Q8 above. Choose the correct option.

1 point

- ☐ ∇f is continuous everywhere
- ☐ ∇f is discontinuous at the origin $(0, 0)$
- ☐ $\frac{\partial f}{\partial x}$ is continuous at the origin but $\frac{\partial f}{\partial y}$ is discontinuous at the origin
- ☐ $\frac{\partial f}{\partial y}$ is continuous at the origin but $\frac{\partial f}{\partial x}$ is discontinuous at the origin
- ☐ Directional derivative of f in the direction of a vector $a \in \mathbb{R}^2$ at $(0, 0)$ cannot be obtained using the gradient of f

No, the answer is incorrect.

Score: 0

Accepted Answers:

∇f is discontinuous at the origin $(0, 0)$
 Directional derivative of f in the direction of a vector $a \in \mathbb{R}^2$ at $(0, 0)$ cannot be obtained using the gradient of f

Your score is: 0/10